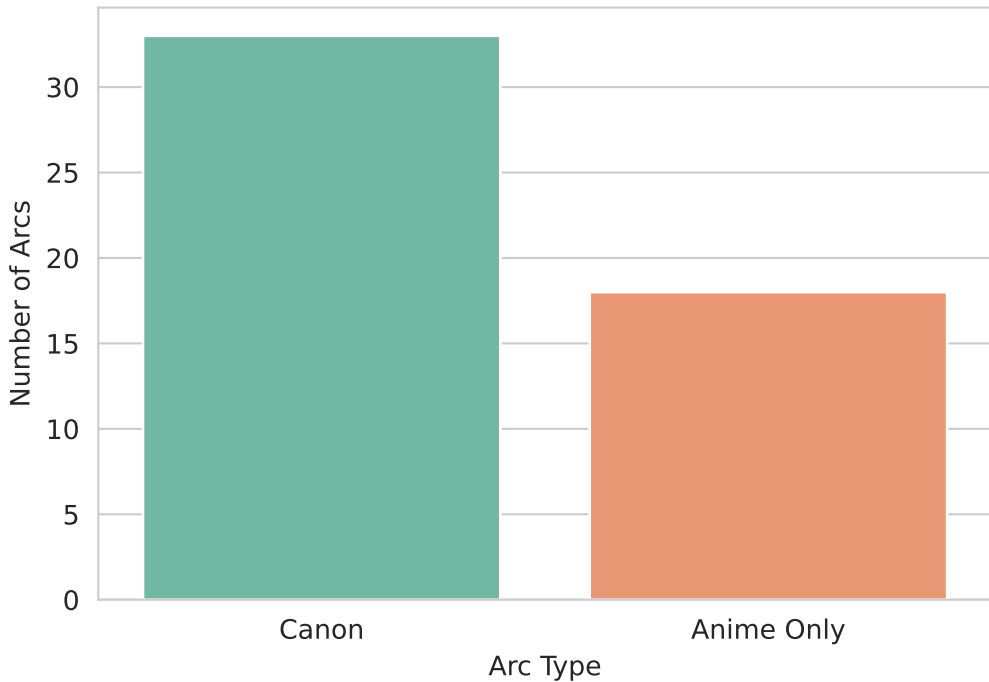
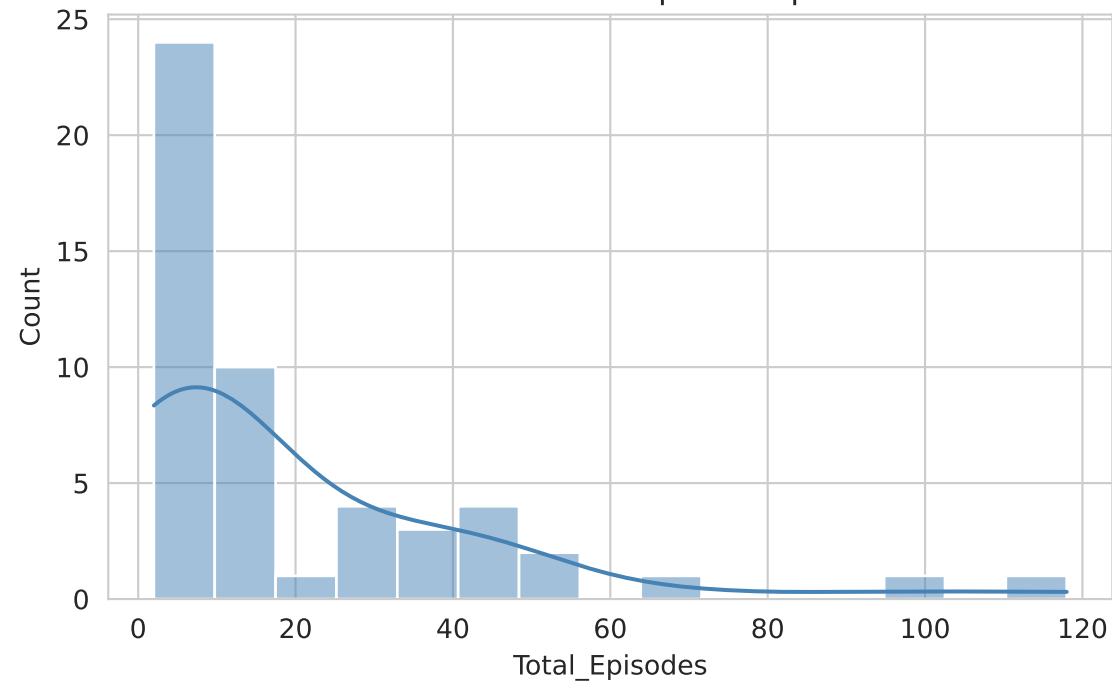


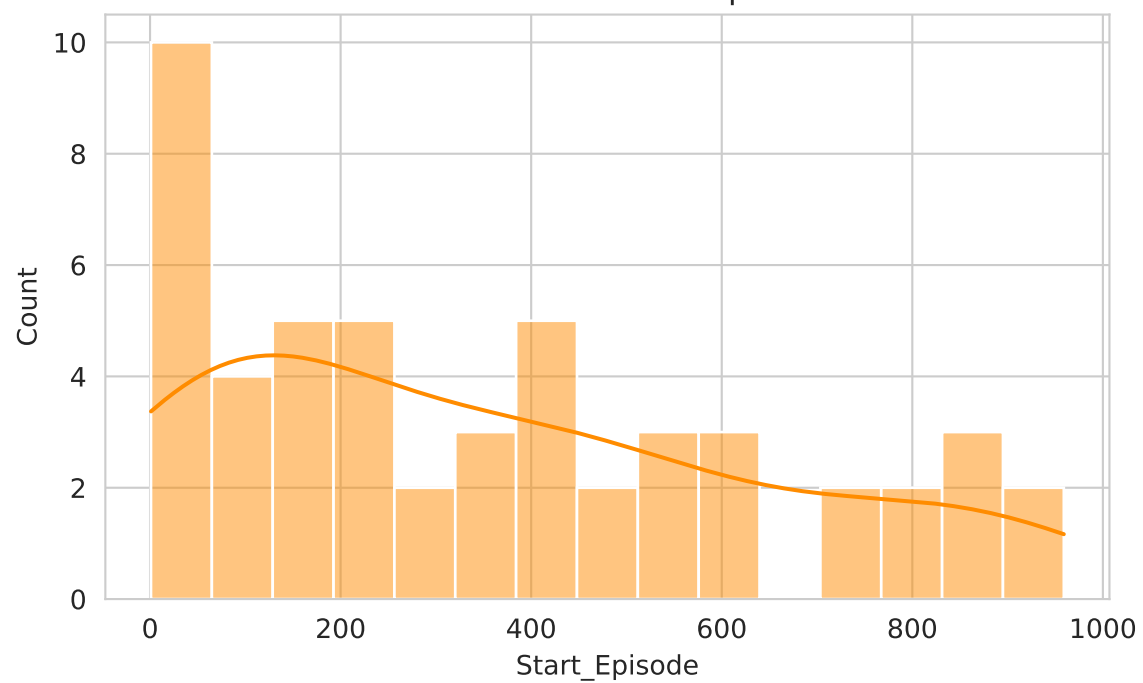
Target Distribution: Canon vs Anime Only Arcs



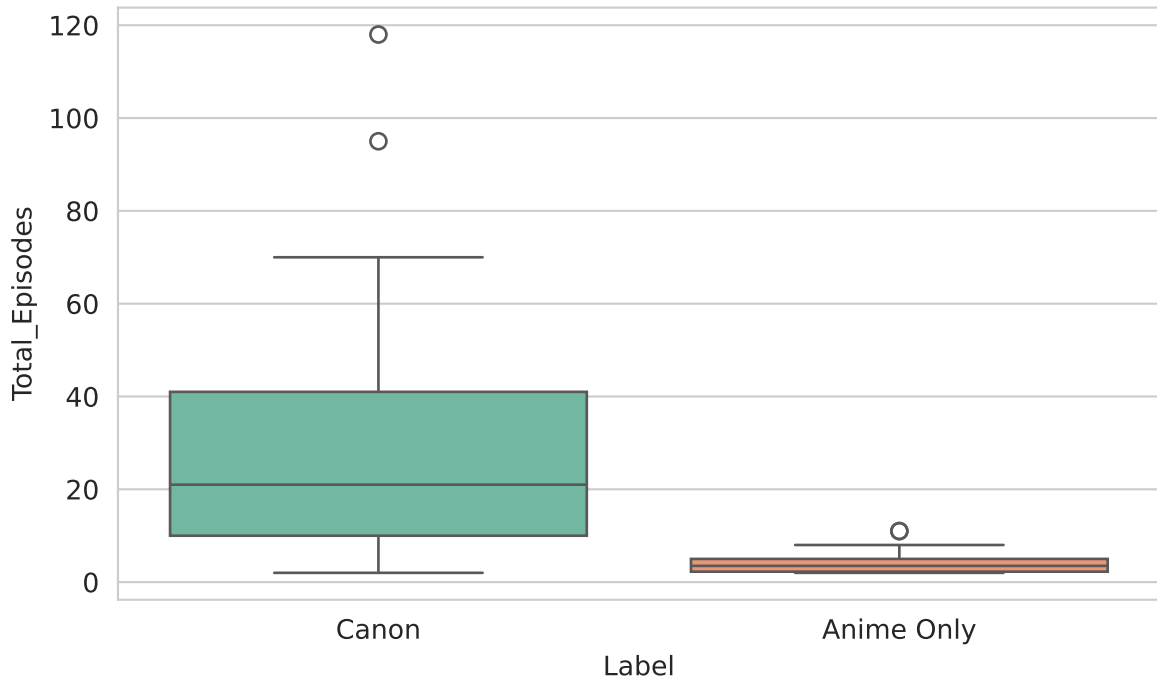
Distribution of Total Episodes per Arc



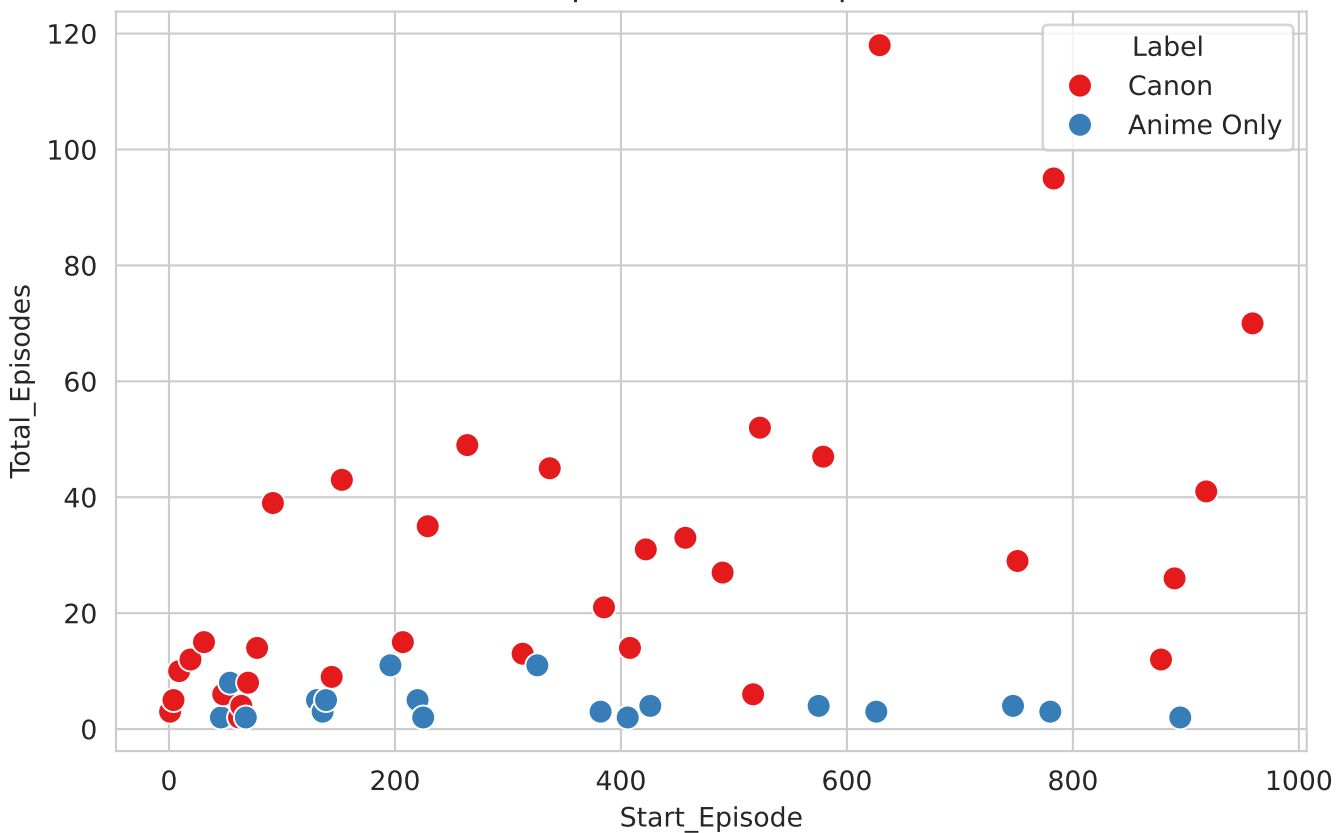
Distribution of Start Episode



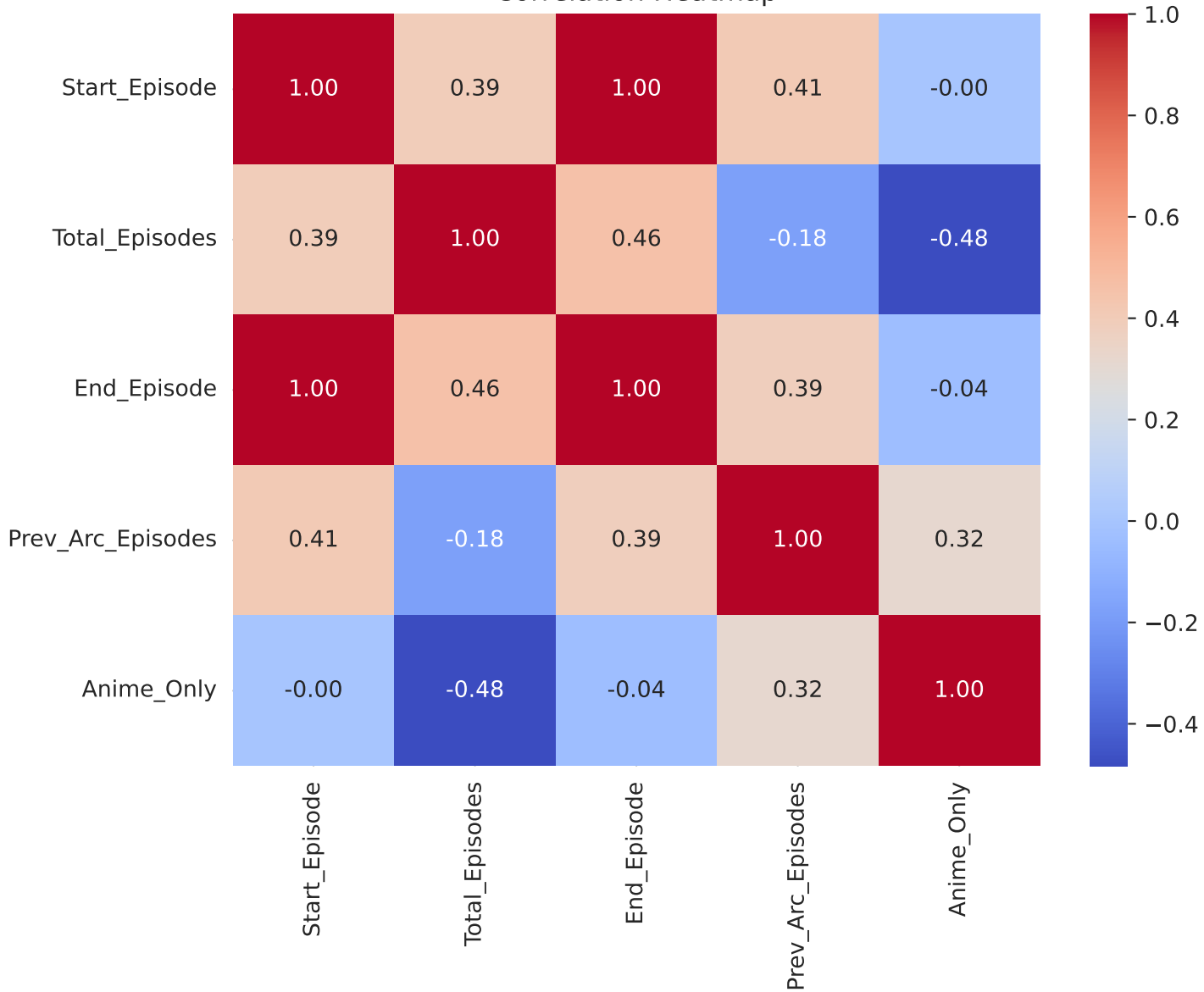
Arc Length by Arc Type



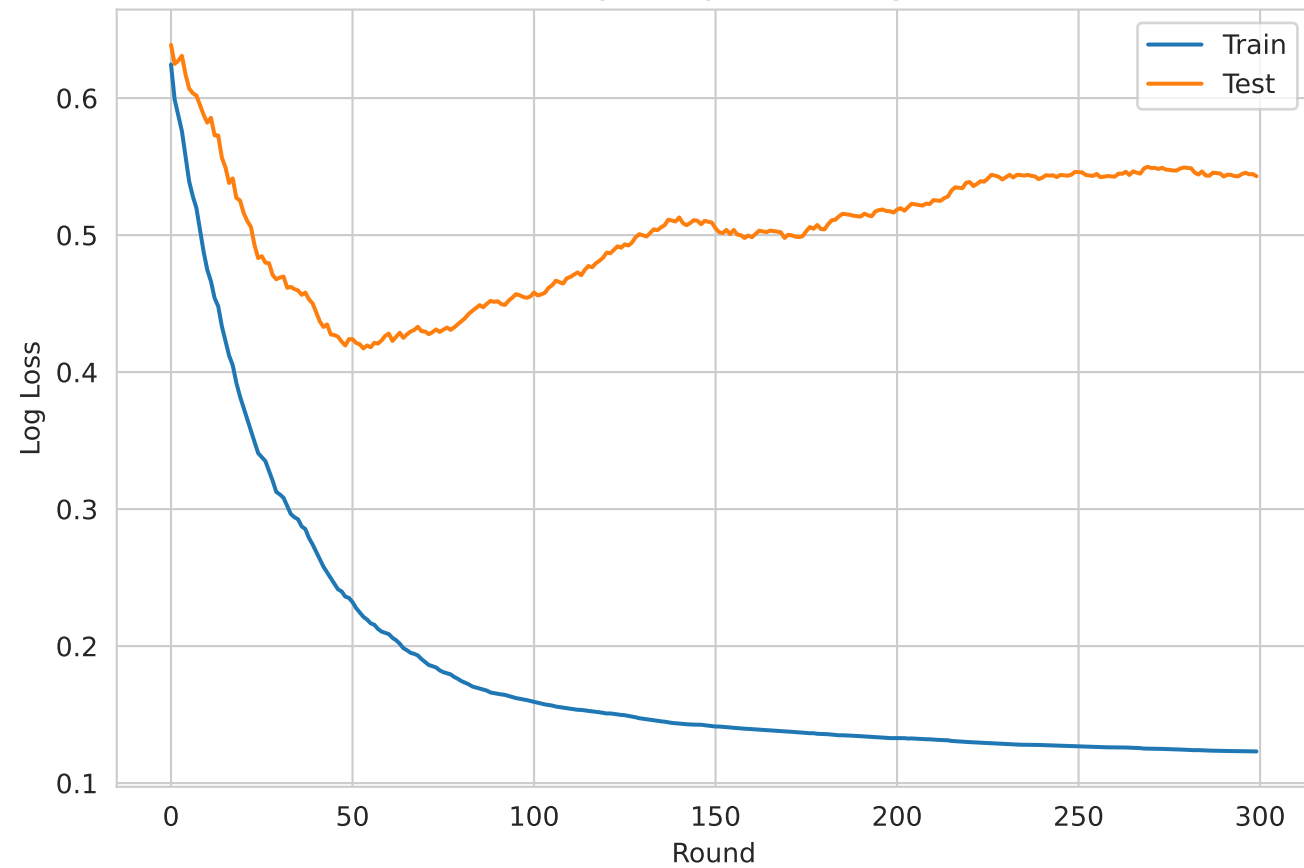
Start Episode vs Total Episodes



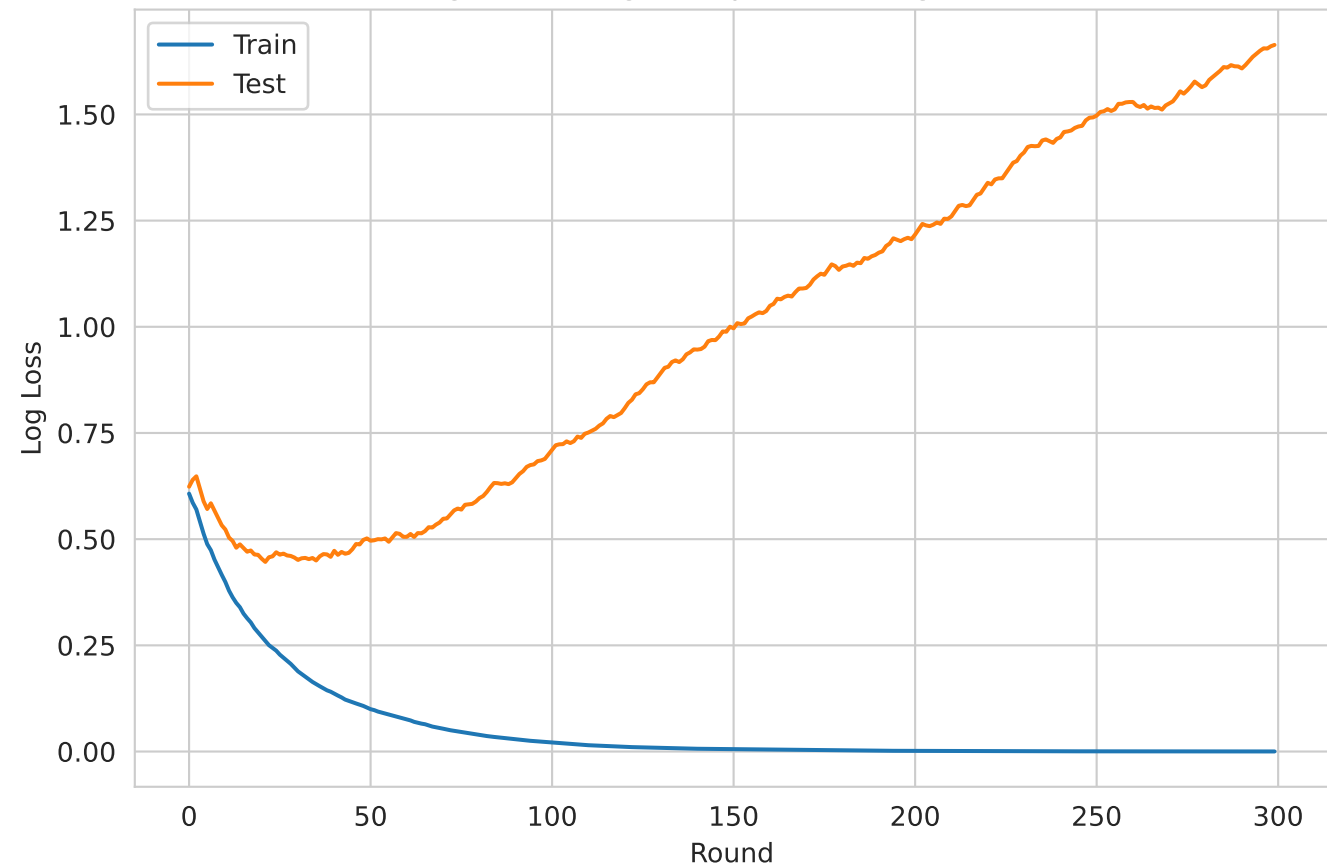
Correlation Heatmap



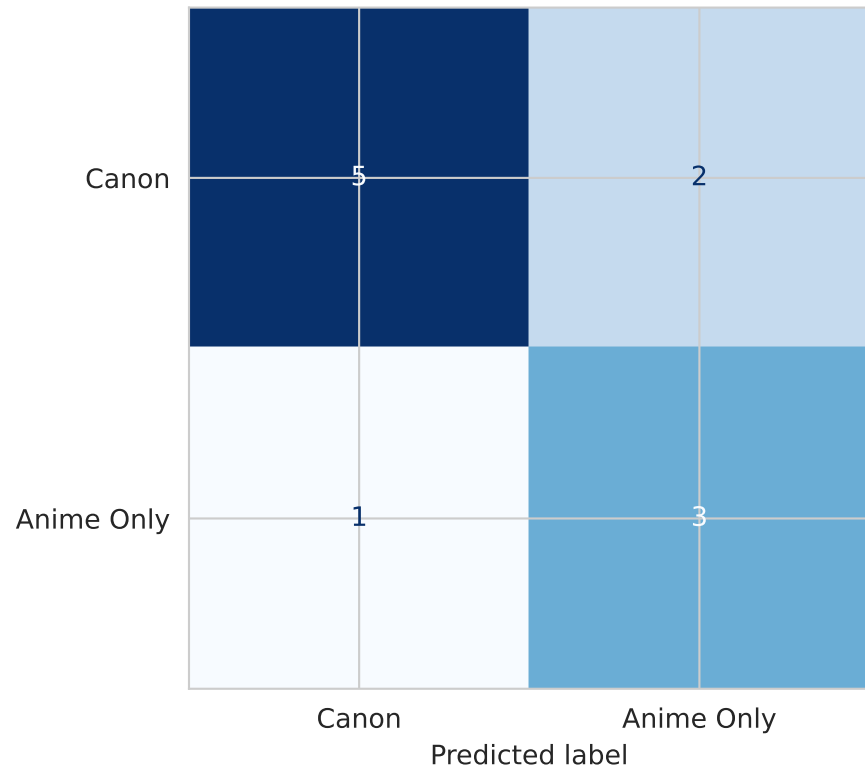
XGBoost Log Loss per Boosting Round



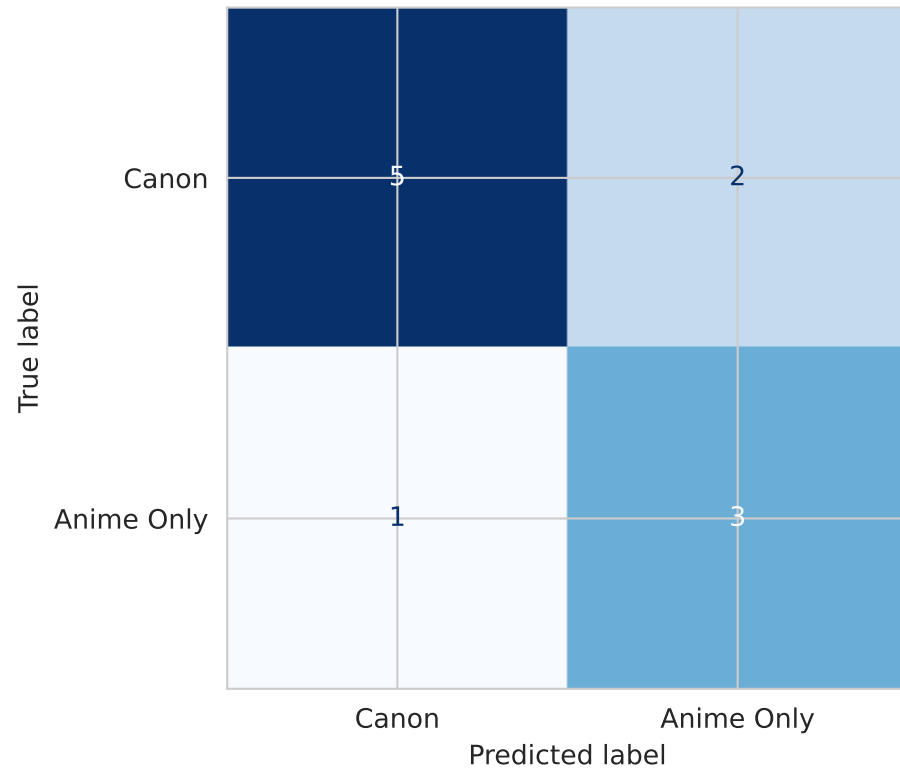
LightGBM Log Loss per Boosting Round



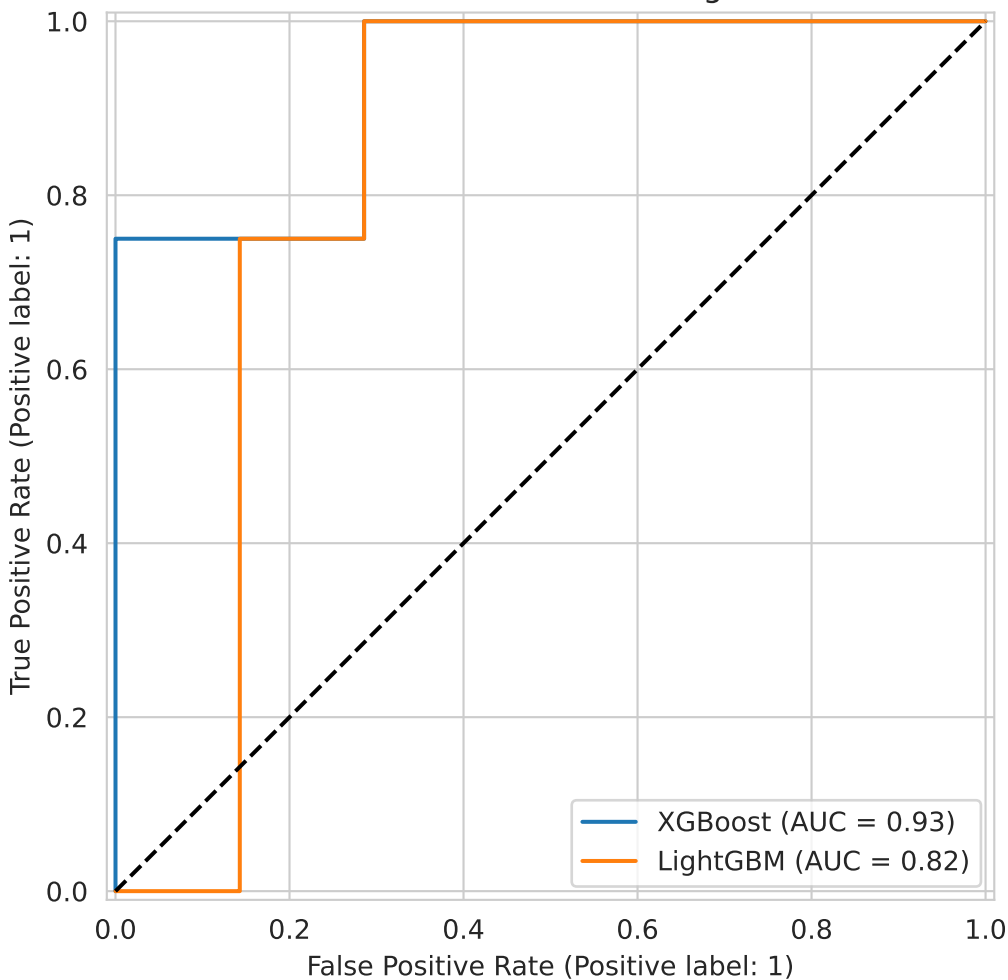
Confusion Matrix: XGBoost



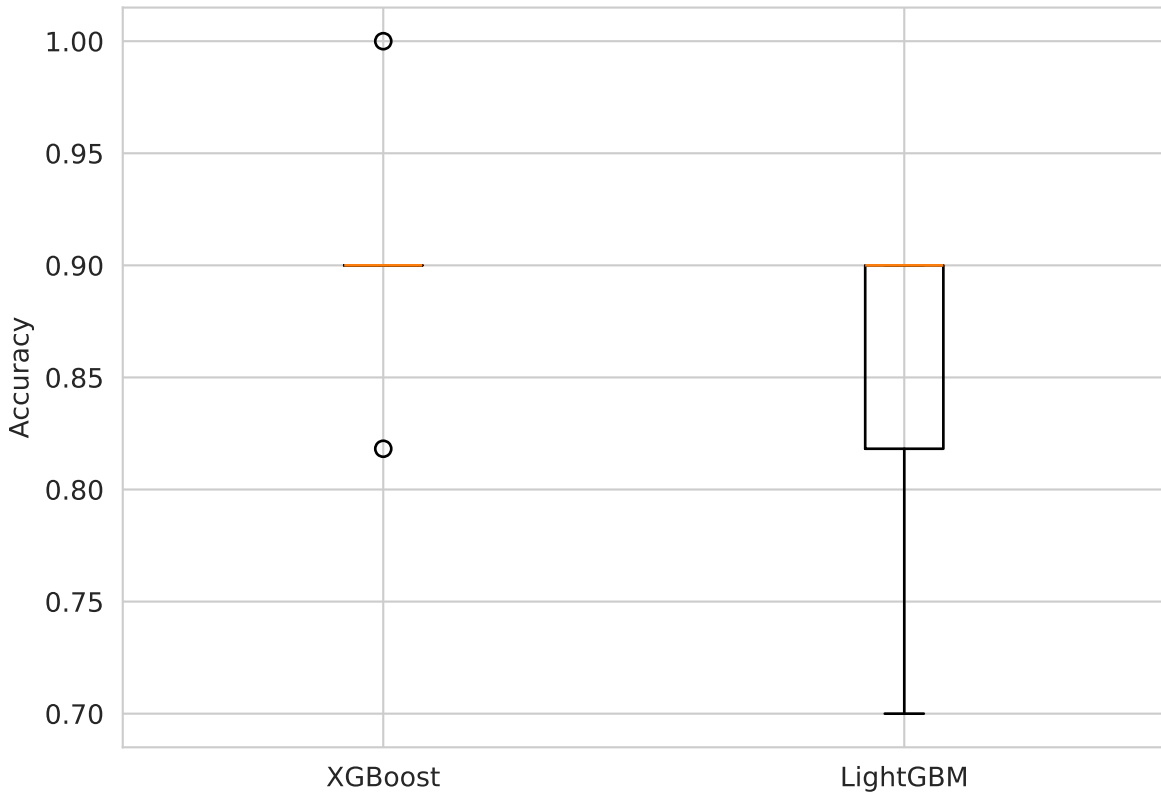
Confusion Matrix: LightGBM



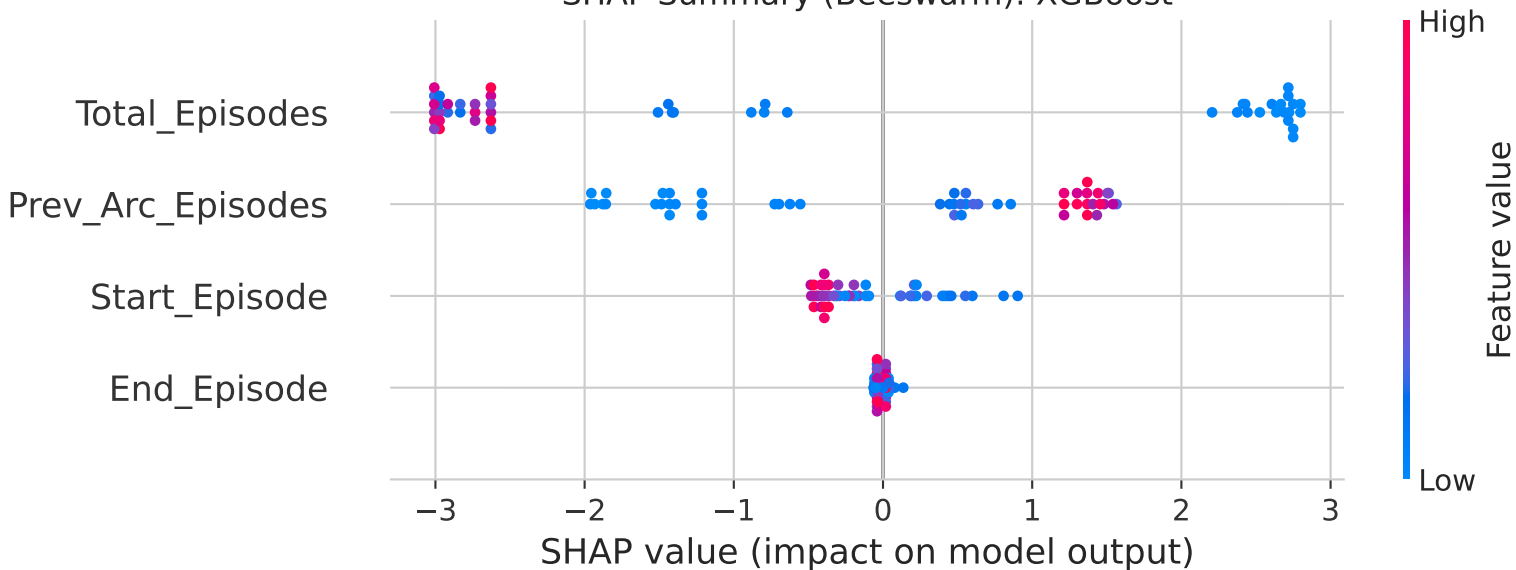
ROC Curves: XGBoost vs LightGBM



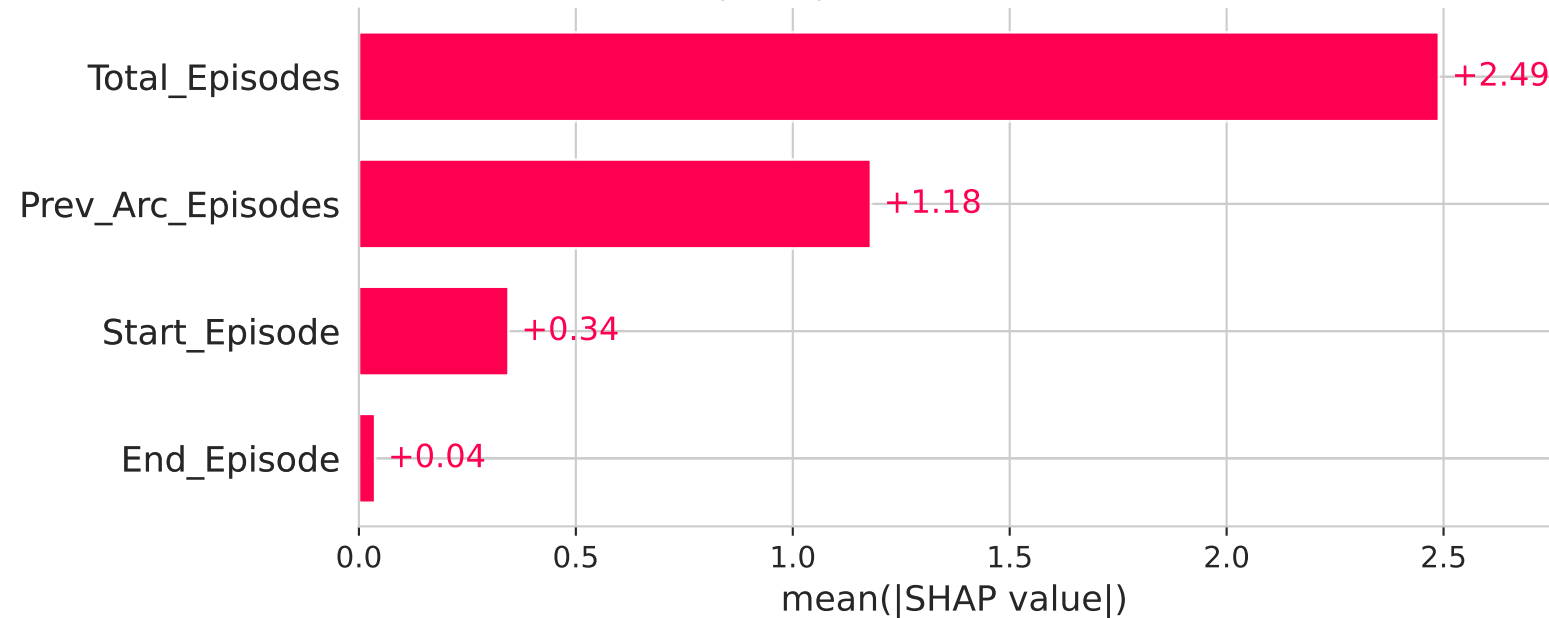
5-Fold CV Accuracy



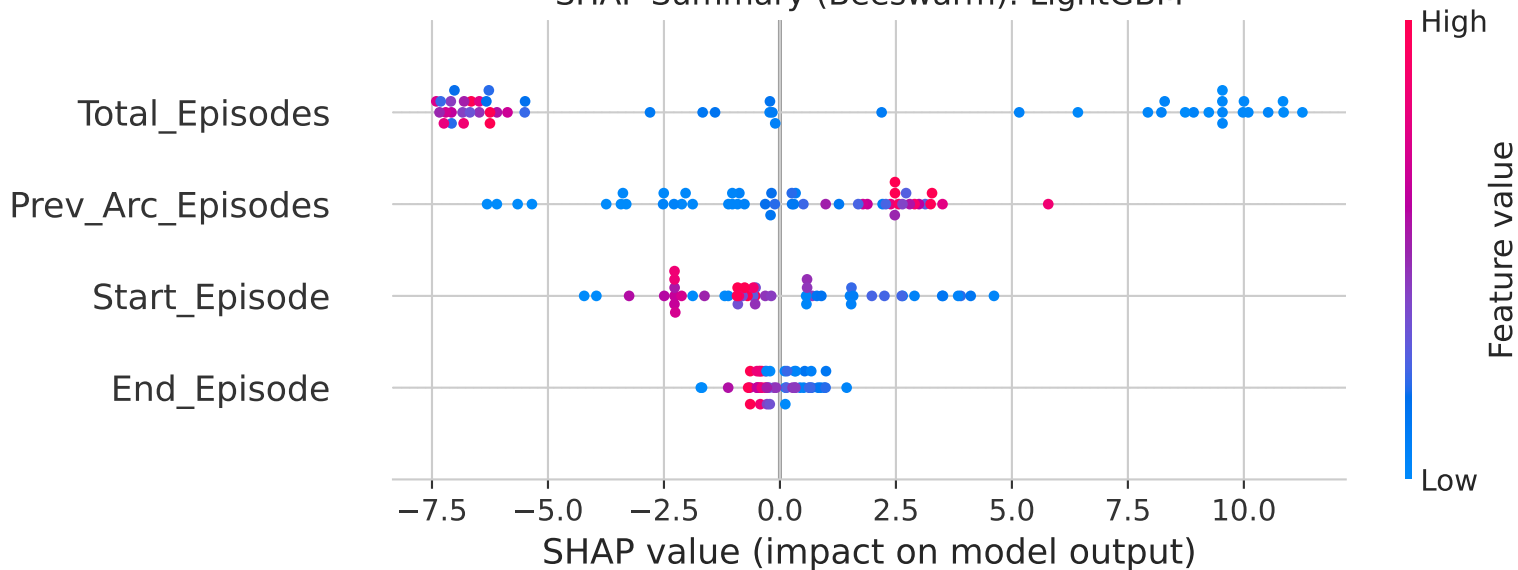
SHAP Summary (Beeswarm): XGBoost



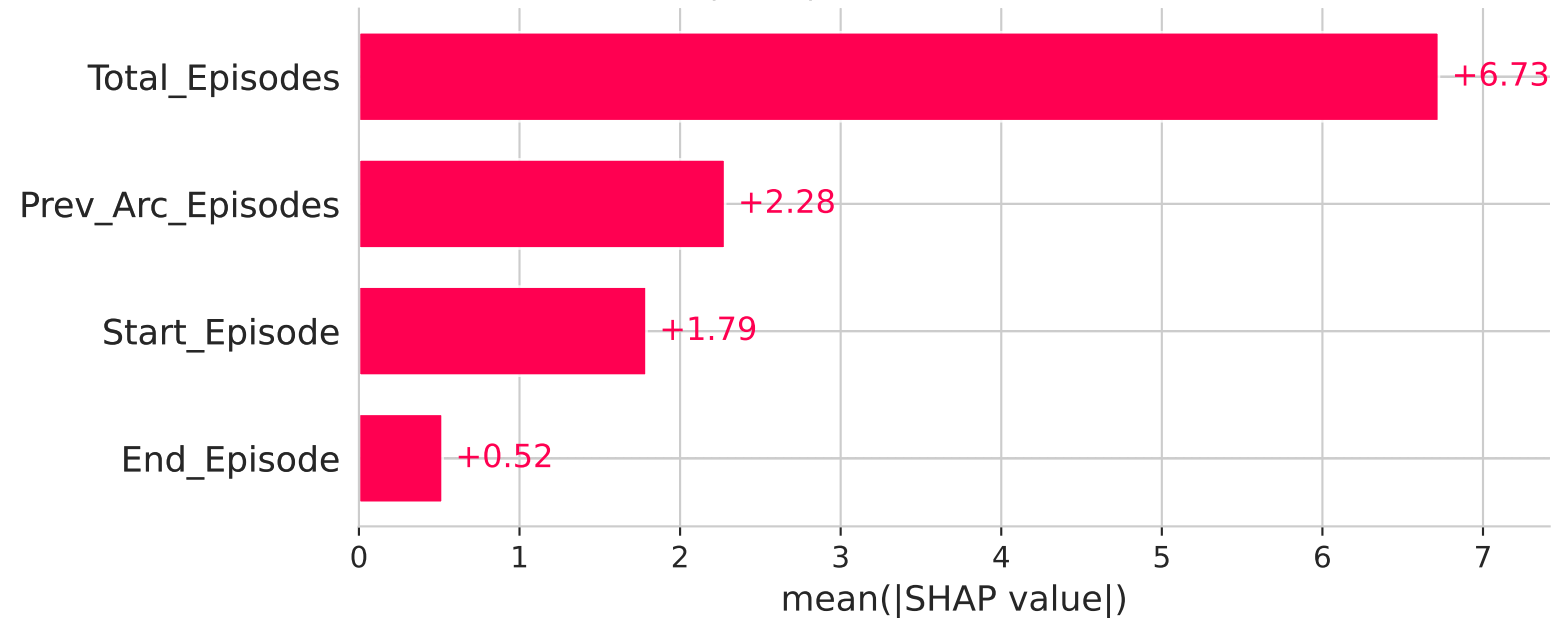
Mean |SHAP| Global Importance: XGBoost



SHAP Summary (Beeswarm): LightGBM



Mean |SHAP| Global Importance: LightGBM



SHAP Waterfall (Single Arc): XGBoost

$f(x) = -5.022$

95 = Total_Episodes

-2.63

3 = Prev_Arc_Episodes

-1.43

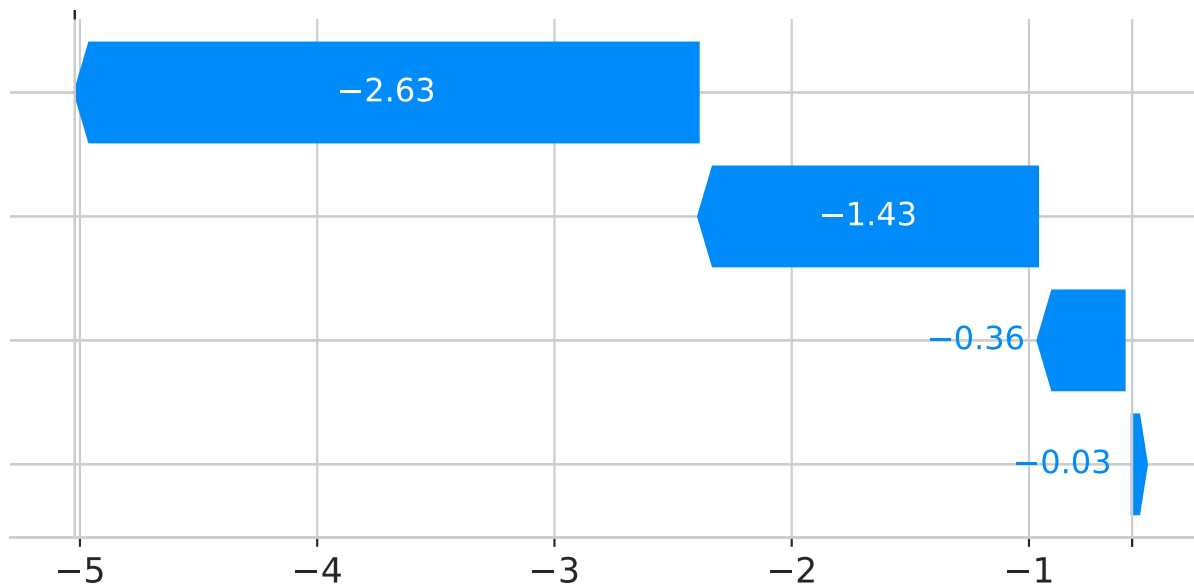
783 = Start_Episode

-0.36

877 = End_Episode

-0.03

$E[f(X)] = -0.566$



SHAP Waterfall (Single Arc): LightGBM

$$f(x) = -13.317$$

95 = Total_Episodes

-6.25

3 = Prev_Arc_Episodes

-2.03

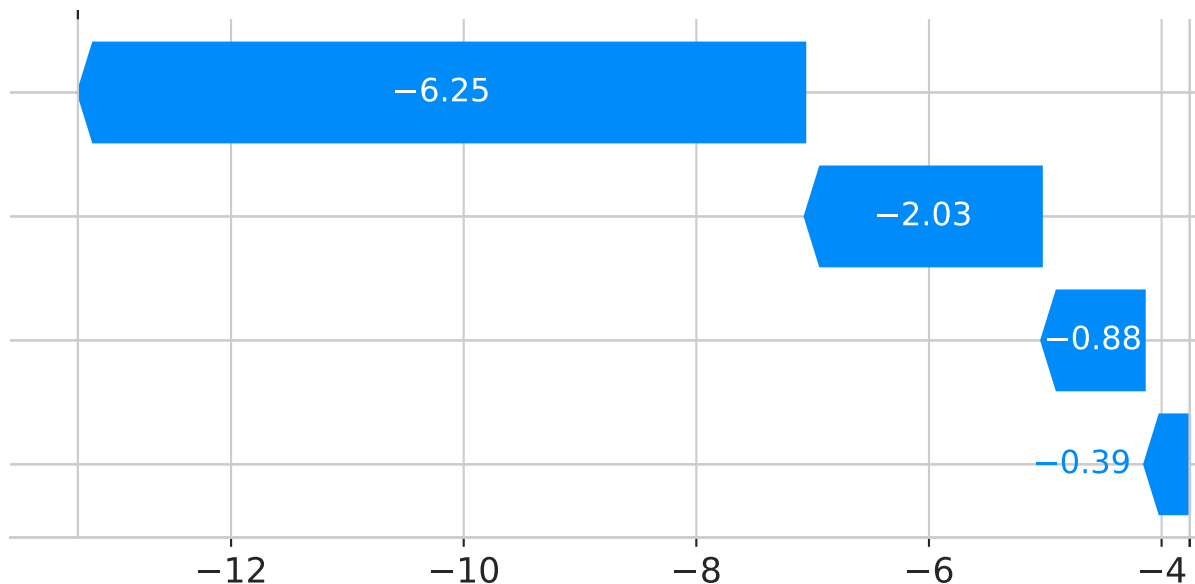
783 = Start_Episode

-0.88

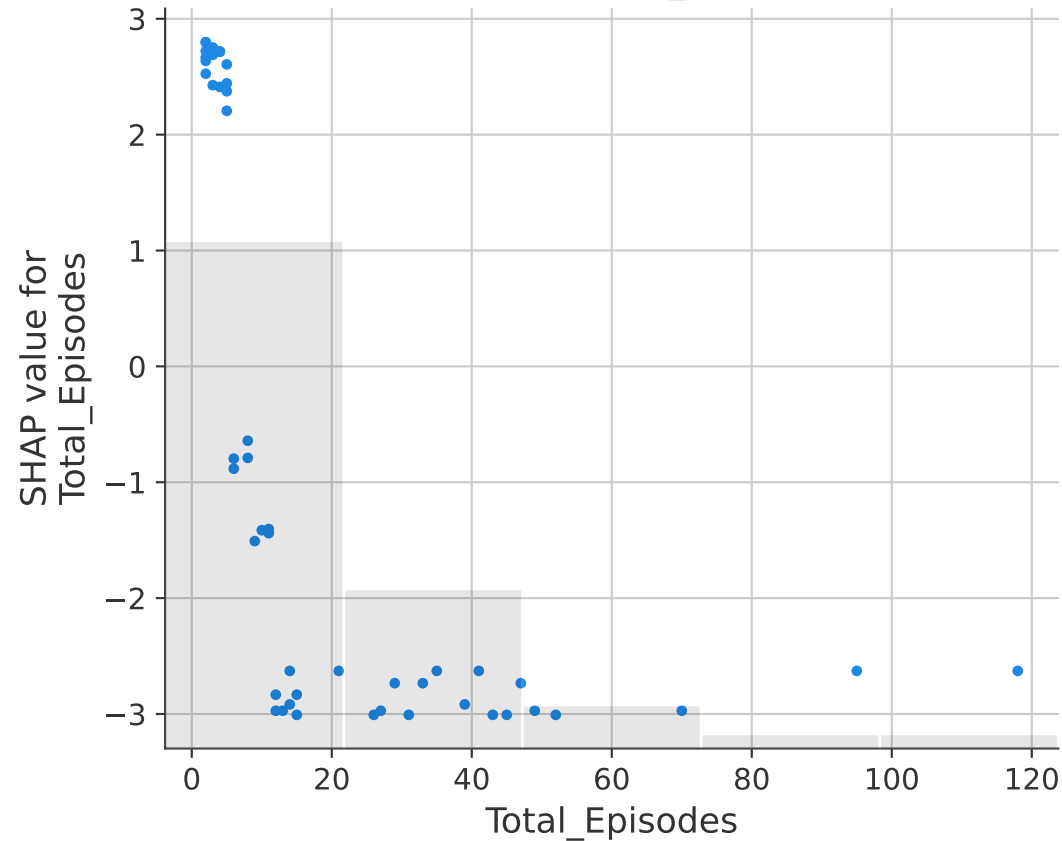
877 = End_Episode

-0.39

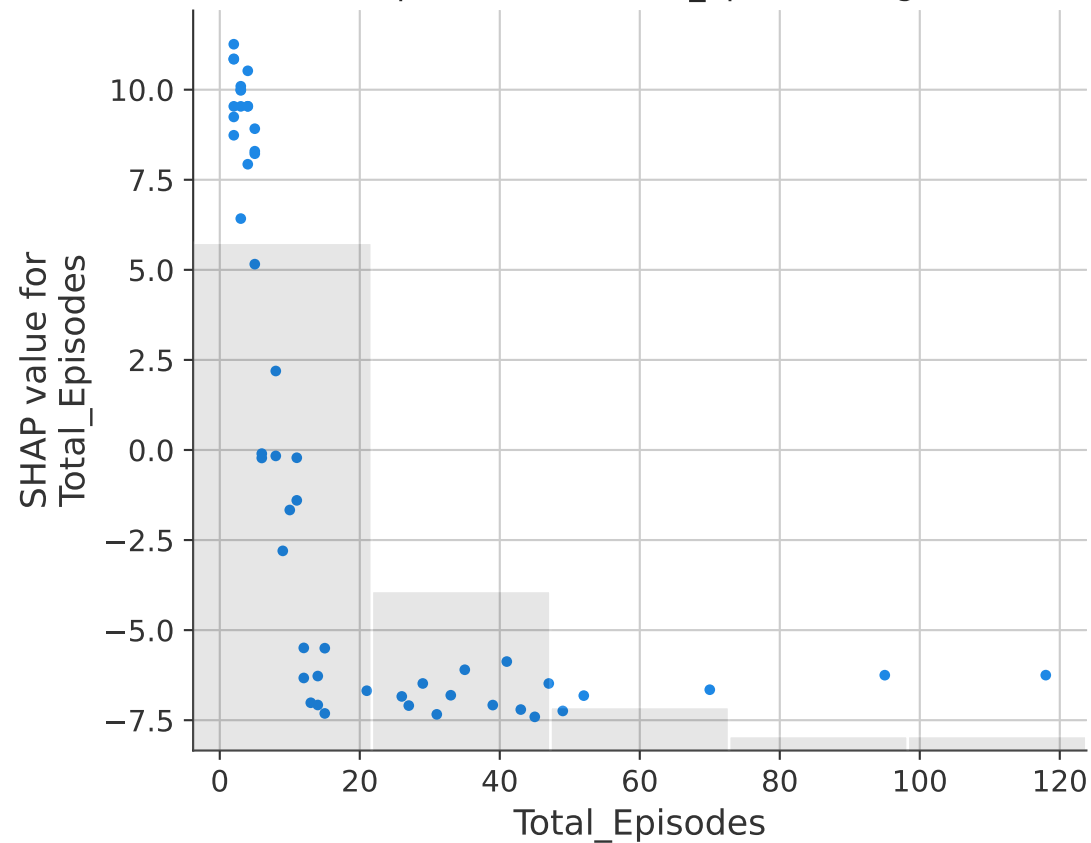
$$E[f(X)] = -3.758$$



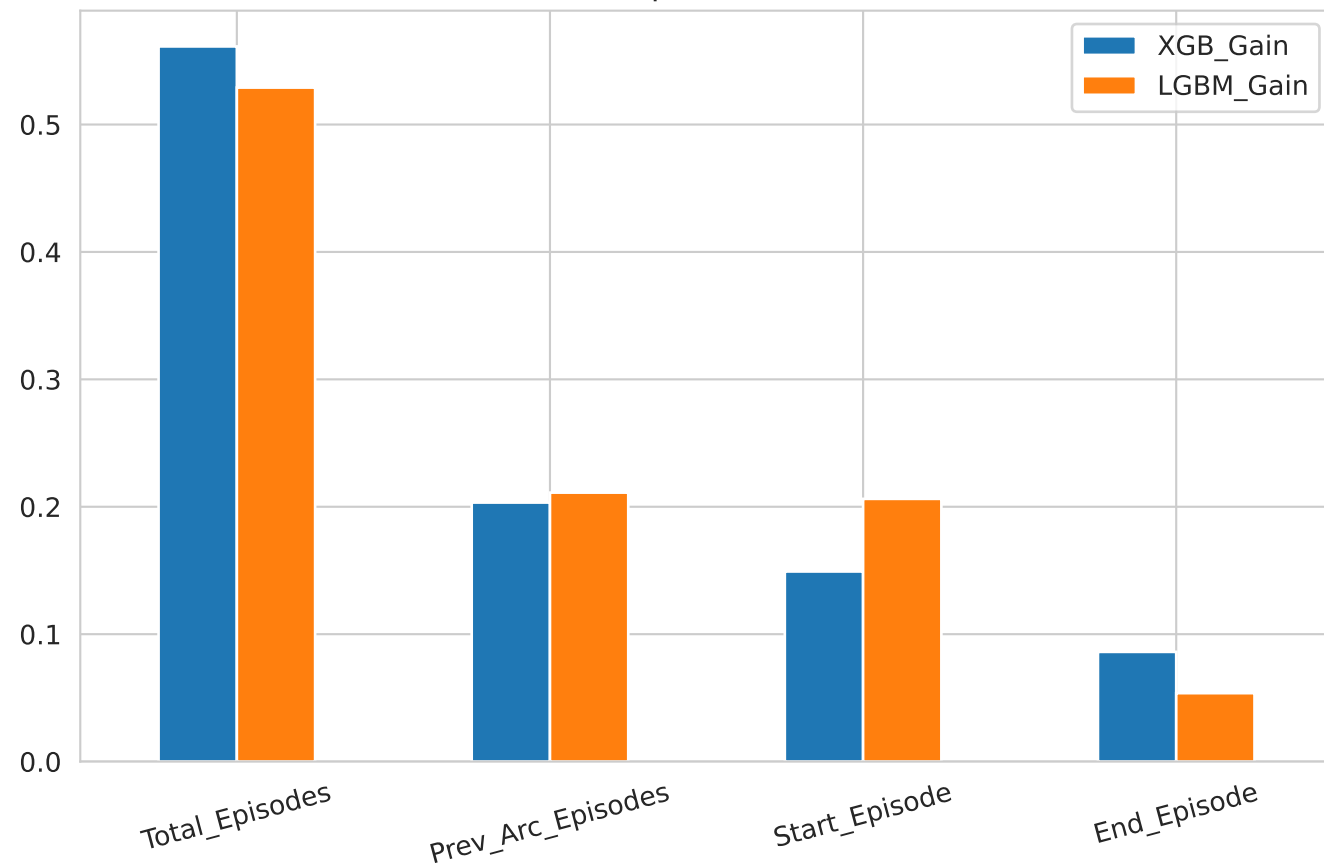
SHAP Dependence on Total_Episodes: XGBoost



SHAP Dependence on Total_Episodes: LightGBM



Built-in Gain Importance (Normalized)



Mean |SHAP| Importance

